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ISOLATION OF NATURAL DYE FROM BEET ROOT AND ITS APPLICATION ON WOOL AND THREAD WITH DIFFERENT MORDANTS AT DIFFERENT TEMPERATURES

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Abstract:

A pre mordanting method was used in dyeing of woolen yarn and thread with natural dye obtained from beet root. Natural dyestuff can produce a wide range of colours by mix and match system. A small variation in the dyeing technique or the use of different mordants with the same dye can shift the colours of a wide range or create totally new colours, which are not easily possible with synthetic dyestuffs.

Natural dyes are usually moth proof and can replace synthetic dyes in kid garments and food stuff for safety.

Introduction:-

Plant dyes appear as interesting alternative because of the health hazards of many synthetic dyes. Active dyestuff parts of plants are used for dyeing. Beet root is known by its red colour. This red colour is because of the betanin pigment found in beetroot. The characteristic feature of this vegetable is it's dark purple skin and a distinctive purple flesh. This betanin pigment is also known as a prominent Anthocyanin antioxidant. Betanin is the main colouring compound present in red beetroot juice colour. The colouring responsible for the red hue of red beet juice is a group of molecules called Betalains. This group of pigments contains the red and yellow pigments known as Betacyanins and Betaxanthins respectively.

Experimental:-

Beetroots(50 g) were cut into small pieces and 450 mL of water was added into it. It was then refluxed on a boiling water- bath for about two hours and was reduced. After the dyebath has cooled, beets pieces were strained from dyebath.

The mordants used in the process were alum, potassium dichromate, vinegar, ammonia and copper sulphate.

At first ,0.2 grams of each of thread and woolen yarn were weighted to be dyed. Each of the weighted thread and wool were soaked in mordant-dye mixture and were kept on the boiling water-bath for about half an hour with constant stirring.

MORDANTING-DYEING PROCESS

The different mordants were taken in different containers containing dye and kept on the boiling water-bath. The weighted wool and thread were immersed into the mordant-dye mixture. In the first step, the immersed wool and thread were kept on the water-bath for heating at a temperature 91⁰C-93⁰C for about half an hour with constant stirring. And in the second step, the immersed wool and thread were boiled on a mantle for about half an hour at a temperature >100⁰C with constant stirring. The dyed wool and thread samples were then rinsed and dried in shade.

PREPARATION OF MORDANTS:

Sl. No.	Mordants	Measurements		Amount of water	Mordant observation
		I	II		
1.	Copper sulphate	0.1 g	0.4 g	40 mL	Light blue
2.	Vinegar	2.5 mL	5 mL	30 mL	Clear liquid
3.	Ammonia	1 mL with 2 mL H ₂ O	2 mL with 2 mL H ₂ O	30 mL	Clear liquid
4.	Alum	3.5 g		30 mL	Clear liquid
5.	Potassium dichromate	0.1 g	0.4 g	40 mL	Orange

MORDANTS WITH DYE OBSERVATIONS :

Sl.No.	Mordants	Volume of dye	Quantity	Observation
1.	Copper sulphate	40 mL	0.1 g	Turned dye greenish brown during heating.
			0.4 g	Became very dark green.
2.	Vinegar	30mL	2.5 mL	No change in colour (dye colour)
			5 mL	No change in colour (dye colour)
3.	Ammonia	30 mL	1 mL with 2 mL H ₂ O	Turned brown
			2 mL with 2 mL H ₂ O	Turned very dark brown almost black
4.	Alum	30 mL	3.5 g	No change in colour (dye colour)
5.	Potassium dichromate	40 mL	0.1 g	Turned light brown
			0.4 g	Turned dark brown

COLOUR OF WOOL AND THREAD AFTER MORDANT-DYE PROCESS:

The mixture of dye with different mordants gives the following colour on wool and thread:

Sl no	Sample	Temperature 91-93 °C (on water bath)	Wool	Thread	Temperature above 100°C (on mantle)	Wool	Thread
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SET-I (Temp: 91-93°C)

Sl.No.	1	2	3	4	5	6	7	8	9
Mordants	CuSO ₄ (0.1g)	CuSO ₄ (0.4g)	Vinegar (2.5mL)	Vinegar (5mL)	Ammonia (1mL)	Ammonia (2 mL)	K ₂ Cr ₂ O ₇ (0.1 g)	K ₂ Cr ₂ O ₇ (0.4 g)	Alum (3.5 g)
Wool	Spring Bud	Android green	Pastel Orange	Pastel Orange	Arylide Yellow	Arylide yellow	Copper	Copper	Indian yellow
Thread	Chartreuse	Pear	Earth Yellow	Earth Yellow	Sandune	Sandune	Cinnamon	Cinnamon	bronze

SET-II (Temp: above 100°C)

Sl.No.	1	2	3	4	5	6	7	8	9
Mordants	CuSO ₄ (0.1g)	CuSO ₄ (0.4g)	Vinegar (2.5mL)	Vinegar (5mL)	Ammonia (1mL)	Ammonia (2 mL)	K ₂ Cr ₂ O ₇ (0.1 g)	K ₂ Cr ₂ O ₇ (0.4 g)	Alum (3.5 g)
Wool	Olive Green	Dark Olive Green	Baby pink	Baby Pink	Seashell	champagne	Sepia	Sepia	Dark salmon
Thread	Pistachio	Asparagus	Pastel Red	Pastel Red	peach	sunset	Golden Brown	Brown	Atomic Tangerine

COMPARISON OF SHADES AT DIFFERENT TEMPERATURES

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1.	Vinegar (2.5 mL)		Pastel orange	Earth yellow		Baby pink	Pastel red
2.	Vinegar (5 mL)		Pastel orange	Earth yellow		Baby pink	Pastel red
3.	Pure dye		Indian yellow	Apricot		Baby pink	Pastel red
4.	Ammonia (1 mL)		Arylide yellow	Sandune		Sea-shell	Peach
5.	Ammonia (2 mL)		Arylide yellow	Sandune		champagne	Sunset
6.	Alum		Indian yellow	Bronze		Dark salmon	Atomic tangerine
7.	Copper sulphate (0.1 g)		Spring bud	Chartreuse		Olive green	pistachio
8.	Copper sulphate (0.4 g)		Android green	Pear		Dark olive green	Asparagus

							
9.	Potassium dichromate (0.1 g)		Copper	Cinnamon		Sepia	Golden brown

SET-I (The DYED THREAD AND WOOL)



Vinegar (2.5 mL)



Vinegar (5 mL)



Pure dye



Alum



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Ammonia (2 mL)



Ammonia (1 mL)



CuSO₄ (0.1 g)



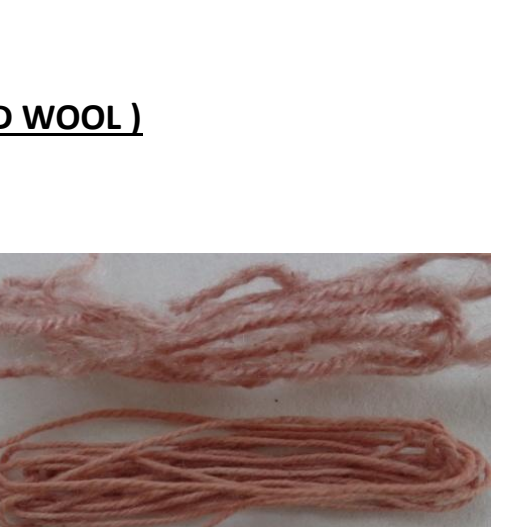
CuSO₄ (0.4 g)



K₂Cr₂O₇ (0.1 g)



K₂Cr₂O₇ (0.4 g)



SET-II (DYED THREAD AND WOOL)



Vinegar (2.5 mL)



Vinegar (5 mL)



Pure dye



Alum



Ammonia (1 mL)



Ammonia (2 mL)



CuSO₄ (0.1 g)



CuSO₄ (0.4 g)



K₂Cr₂O₇(0.1 g)



K₂Cr₂O₇(0.4 g)

ADVANTAGES OF NATURAL DYES

- (1) Natural dyes are less toxic than their synthetic counterpart. Many of the natural dyes like turmeric, annatto and saffron are permitted as food additives. Many natural dyes have pharmacological effects and possible health benefits.
- (2) The shades produced by natural dyes are usually soft, lustrous and soothing to the human eye.
- (3) Natural dyestuff can produce a wide range of colours by mix and match system. A small variation in the dyeing technique or the use of different mordants with the same dye can shift the colours of a wide range or create totally new colours, which are not easily possible with synthetic dyestuffs.
- (4) Unlike non-renewable basic raw materials for synthetic dyes, the natural dyes usually renewable being agro-renewable /vegetable based and at the same time biodegradable.
- (5) Natural dyes are usually moth proof and can replace synthetic dyes in kid garments and food-stuffs for safety.
- (6) Some of the natural dyes are enhanced with age, synthetic dyes fade with times.
- (7) In some cases like harda, indigo etc., the waste in the process becomes ideal fertilizers for use in agricultural fields. Therefore no disposal problem in this natural waste.
- (8) Many plants thrive on wastelands. Thus, wasteland utilization is an added merit of the natural dyes. Dyes like madder grows as host in tea gardens. So there is no additional cost or effort to grow it.
- (9) This is a labour intensive industry, thereby providing job opportunities for all those engaged in cultivation, extraction and application of these dyes on textile/ food/ leather etc.
- (10) Application of natural dyes has potential to earn carbon credit by reducing consumption of fossil fuel(petroleum) based synthetic dyes.
- (11) Natural dyes bleed but do stain on fabric, turmeric being an exception.

DISADVANTAGES OF NATURAL DYES

- (1) It is difficult to reproduce shades by using natural dyes/colourants, as these agroproduct vary from one crop season to another crop season, place to place and species to species, maturity period etc.
- (2) It is difficult to standardize a recipe for the use of natural dyes as the natural dyeing processes and its colour development depends not only on colour components but also on materials.
- (3) Natural dyeing requires skilled workmanship and is therefore expensive. Low colour of source natural dyes thus necessitates the use of more dyestuffs, larger dyeing time and excess cost for mordants and mordanting.
- (4) Scientific backup of a large part of the science involved in natural dyeing is still need to be explored.
- (5) Lack of availability of precise technical knowledge on extraction and dyeing techniques.
- (6) The dyed textile may change colour when exposed to the sun, sweat and air.

CONCLUSION:

It is observed that the beetroot is not only used as food stuff but it can also be used as natural dye as it contain betacyanin which is responsible to impart red colour. Unlike thread (cotton), wool is highly receptive towards mordants and therefore the differences in the colours of wool and thread were observed after dyeing with the same mordants. Wool is softer than the thread and hence absorbed dye easily as compared to thread. The different shades were observed with different mordant at different temperature and time. It also depends on the amount of mordant and dye.

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